WIRELESS COMMUNICATION NETWORKS AND SYSTEMS

LAB 02 REPORT

Transmission Impairments and Noise

Python simulation • ***BELHIMER Mohmaed Ayoub***

1. EXPERIMENTAL RESULTS & ANSWERS TO QUESTIONS

Step 1 — Reference Signals (Clean)

We generate two clean baseline signals at `f0 = 200 Hz` (`fs = 10000 Hz`, `T = 0.05 s`): a single-tone sine wave and a digital-like square wave. We plot the first 10 ms in time alongside their one-sided magnitude spectrum (RFFT).

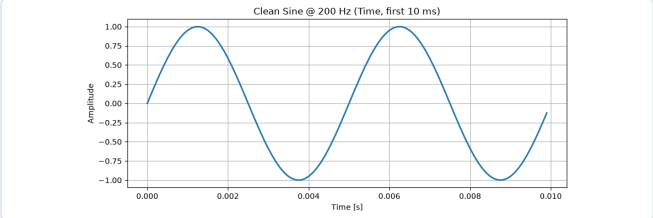


Figure 1a: Clean 200 Hz sine wave (time, first 10 ms).

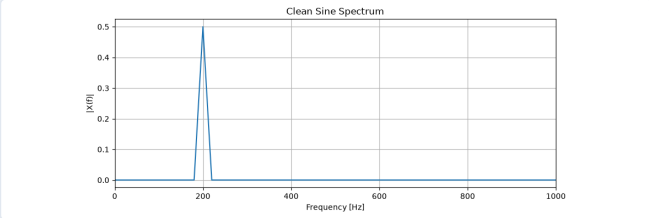


Figure 1b: Clean sine one-sided magnitude spectrum.

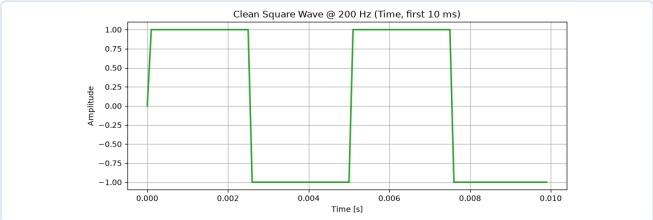


Figure 2a: Clean 200 Hz square wave (time, first 10 ms).

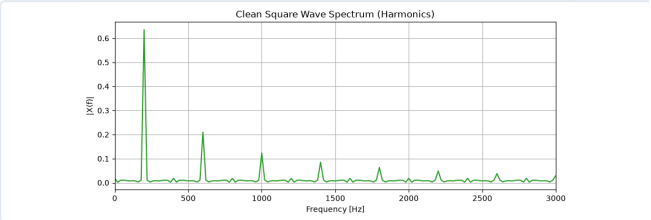


Figure 2b: Square wave spectrum showing odd harmonics.

> **Observation:** The sine wave concentrates its energy at the fundamental frequency (200 Hz). The square wave occupies much more bandwidth due to an infinite series of odd harmonics (200 Hz, 600 Hz, 1000 Hz, 1400 Hz, …), which are required to construct sharp transitions.

Step 2 — Attenuation

We simulate path loss by scaling signal amplitude: `sine_att = 0.3 * sine`. Both time-domain waveform and spectrum are compared against the clean signal.

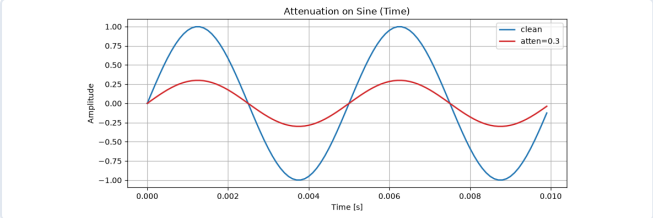


Figure 3a: Attenuation on sine (time, clean vs. attenuated).

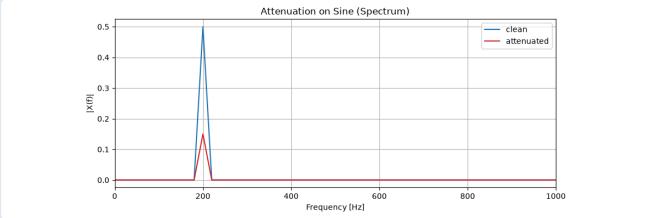


Figure 3b: Attenuation on sine (spectrum, clean vs. attenuated).

> **Key Point:** Attenuation reduces amplitude (and total signal power) without changing spectral shape or creating new frequency components.

Step 3 — Distortion (Limited Bandwidth)

To simulate limited transmission bandwidth, we apply an M-point moving average low-pass filter (`M = 31`) to the square wave, mimicking high-frequency cutoff.

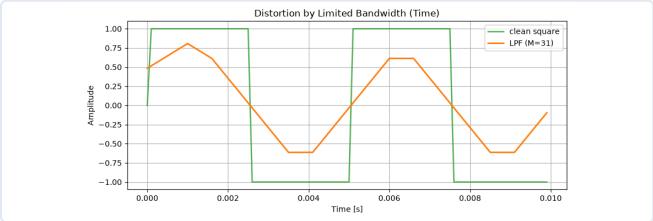


Figure 4a: Bandwidth-limited square wave (time, rounded edges).

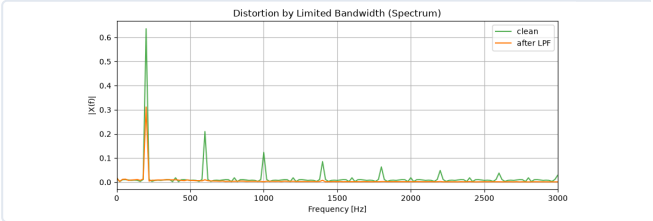


Figure 4b: Spectrum after LPF (high harmonics attenuated).

> **Key Point:** Distortion suppresses higher-order harmonics, rounding the sharp transitions and substantially altering symbol shapes.

Step 4 — Delay and Simple Multipath

We simulate a two-path multipath channel with a direct ray (gain $a_1 = 1.0$) and a delayed echo (gain $a_2 = 0.5$, delay $D = 1$ ms = 10 samples). We observe the result on both sine and square waves.

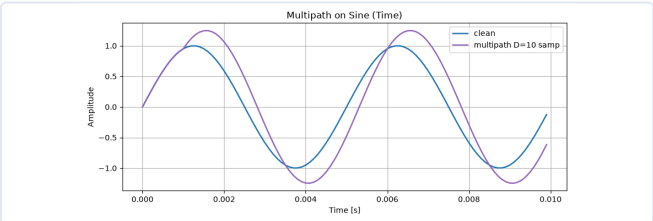


Figure 5a: Multipath on sine wave (amplitude & phase ripple).

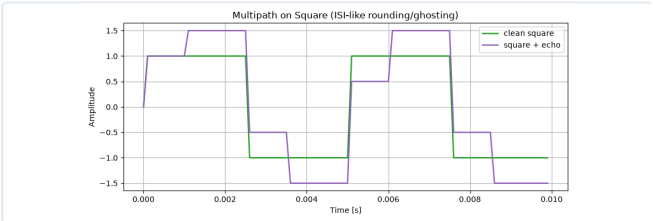


Figure 5b: Multipath on square wave (ISI-like stepping/ghosting).

> **Key Point:** Multipath generates delayed replicas that produce constructive and destructive interference, leading to severe Inter-Symbol Interference (ISI) in digital signals.

Step 5 — Additive White Gaussian Noise (AWGN) and SNR

We add AWGN to achieve a baseline target of `SNR = 6 dB` (measured ~ `5.99 dB`) to observe noise in both domains.

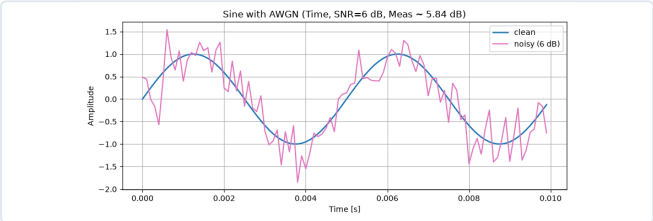


Figure 6a: Sine with AWGN at target SNR = 6 dB (Time domain).

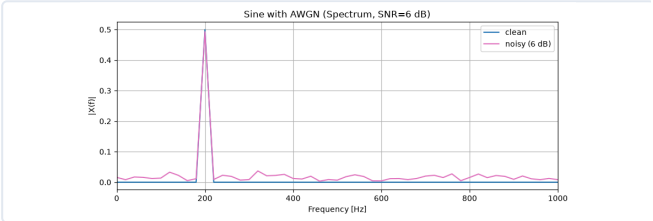


Figure 6b: Spectrum at SNR = 6 dB (Tone peak rising above noise).

> **Key Point:** AWGN adds random zero-mean Gaussian fluctuations across all time samples and distributes its power evenly across all frequency bins.

Step 5 (Continued) — Mini-Exercise: SNR Comparison (0 dB vs. 12 dB)

We repeat AWGN injection across multiple signal-to-noise ratios (0 dB, 6 dB, 12 dB) to evaluate tone peak emergence.

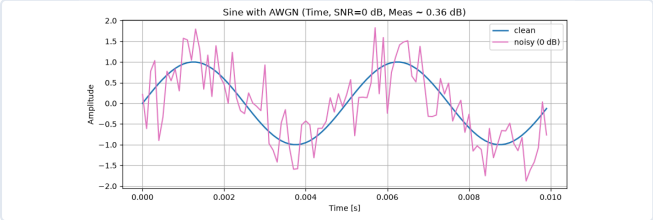


Figure 7a: Sine with AWGN at SNR = 0 dB (Time domain).

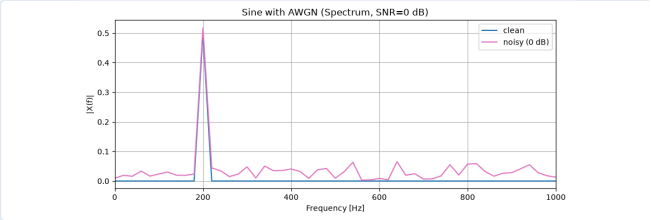


Figure 7b: Spectrum at SNR = 0 dB (High noise floor).

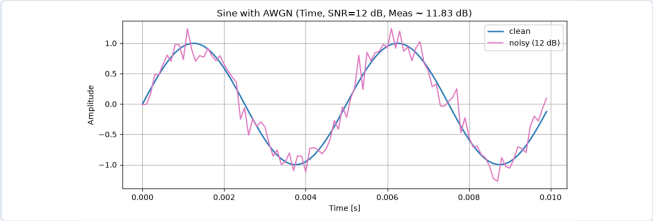


Figure 8a: Sine with AWGN at SNR = 12 dB (Time domain).

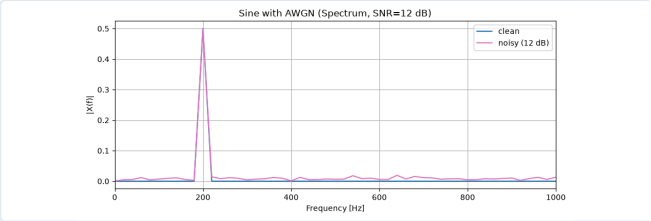


Figure 8b: Spectrum at SNR = 12 dB (Distinct spectral peak).

Mini-exercise: Repeat for SNR = 0, 6, 12 dB and comment how the tone peak emerges above the noise floor.

Observation / Answer: At 0 dB, the noise floor is high and the 200 Hz tone peak is barely distinguishable above the random noise. As SNR increases to 6 dB and 12 dB, the noise floor drops significantly and the tone peak clearly emerges above the noise.

Step 6 — Combined Channel (Attenuation + Distortion + Delay + Noise)

We transmit the digital square wave through a comprehensive channel: attenuation (`A = 0.6`), band-limiting LPF (`M = 21`), multipath echo (`D = 0.7 ms` , a_2 `= 0.4`), and noise (`SNR = 8 dB`).

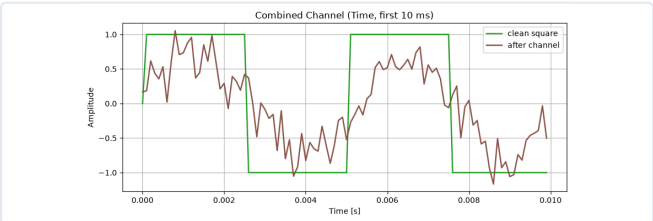


Figure 9a: Combined channel effect (time domain, first 10 ms).

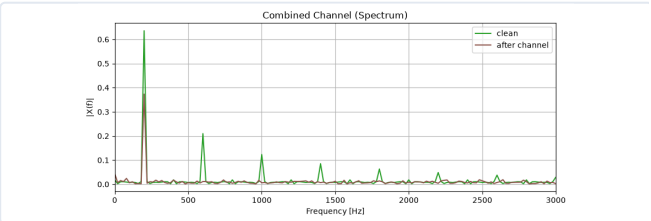


Figure 9b: Combined channel effect (frequency spectrum).

Reflection: Which impairment had the largest visual impact on shape? Which mainly reduced power? How would a symbol detector be affected?

Answer: Limited bandwidth (distortion) and multipath delay had the largest visual impact on shape by rounding sharp edges and adding echo ripples. Attenuation mainly reduced power without altering the waveform shape. A symbol detector would suffer from inter-symbol interference (ISI) and a reduced noise margin, significantly increasing the bit error rate (BER).

DISCUSSION QUESTIONS

Question 1: Which impairment reduces amplitude without changing spectral shape? Why?
Answer: Attenuation. It scales the signal by a constant factor across all frequencies, reducing amplitude and power uniformly without altering the frequency content or relative spectral proportions.

Question 2: Why does limited bandwidth distort square waves more than sines?
Answer: A sine wave consists of a single frequency, so it passes undistorted if within the channel bandwidth. A square wave is composed of a fundamental frequency plus infinite odd harmonics; limited bandwidth attenuates or removes these high-frequency harmonics, rounding the sharp transitions and distorting the waveform.

Question 3: How does multipath cause inter-symbol interference (ISI)?
Answer: Multipath causes delayed copies (echoes) of transmitted symbols to arrive at the receiver over multiple paths. These delayed copies overlap and interfere with subsequent symbols, causing constructive and destructive interference that blurs symbol boundaries and creates ISI.

Question 4: In the spectrum plots with noise, why is the tone still visible at moderate SNR?
Answer: The sinusoidal tone concentrates all its energy into a single narrow frequency bin, producing a sharp spectral peak, while white noise spreads its power evenly across all frequencies. At moderate SNR, this tone peak remains well above the average noise floor.

Question 5: Give a real-world situation where delay is significant (e.g., GEO satellites).
Answer: Geostationary Earth Orbit (GEO) satellite communication. Because satellites orbit at roughly 36,000 km, the round-trip propagation delay is approximately 500 ms, causing noticeable latency in real-time voice calls, video conferencing, and online gaming.

2. CONCLUSION

1. We successfully simulated the four core wireless transmission impairments: attenuation, distortion, multipath delay, and additive noise.
2. Attenuation scales signal amplitude and power uniformly across all frequencies, keeping the spectral shape unchanged.
3. Bandwidth limits act as low-pass filters that cut off higher harmonics, rounding sharp transitions and causing waveform distortion.
4. Multipath propagation creates delayed replicas that interfere constructively and destructively, causing severe Inter-Symbol Interference (ISI).
5. In the frequency domain, narrowband tones remain visible at moderate SNR because signal energy concentrates into a single peak while AWGN spreads over the spectrum.
6. The combined channel illustrates how multiple impairments jointly degrade communications, motivating equalization, filtering, and coding techniques in real wireless systems.